

Exploratory Data Analysis on Severe Weather Event Costs

This project uses real-world storm event data from NOAA to demonstrate fundamental data science techniques in MATLAB. We'll import, clean, and analyze data to understand which severe weather events cause the most economic damage in the U.S.

Data Import Section

```
% Set file path
filename = 'StormEvents_locations-ftp_v1.0_d2023_c20250401.csv';

opts = detectImportOptions(filename);
opts.SelectedVariableNames = {'YEARMONTH', 'EPISODE_ID', 'EVENT_ID',
                              'LOCATION', ...
                              'LATITUDE', 'LONGITUDE', 'LAT2', 'LON2'};
```

```
stormLocations = readtable(filename, opts);

% Preview the dataset
head(stormLocations)
```

YEARMONTH	EPISODE_ID	EVENT_ID	LOCATION	LATITUDE	LONGITUDE	LAT2
2.023e+05	1.7674e+05	1.074e+06	{'ABERNATHY ARPT'}	35.18	-87.05	3.5108e+06
2.023e+05	1.7674e+05	1.074e+06	{'PROSPECT'}	35.01	-87.01	35600
2.023e+05	1.7674e+05	1.074e+06	{'NASHVILLE'}	36.17	-86.81	3.6102e+06
2.023e+05	1.7674e+05	1.074e+06	{'GOODLETTSVILLE'}	36.322	-86.712	3.6193e+06
2.023e+05	1.7674e+05	1.074e+06	{'MILLERSVILLE'}	36.339	-86.71	3.6204e+06
2.023e+05	1.7674e+05	1.074e+06	{'GOODLETTSVILLE'}	36.32	-86.72	3.6192e+06
2.023e+05	1.7674e+05	1.074e+06	{'CORNERSVILLE'}	35.323	-86.869	3.5194e+06
2.023e+05	1.7674e+05	1.074e+06	{'JEFFERSON'}	35.99	-86.48	3.5594e+06

```
summary(stormLocations)
```

```
stormLocations: 43572x8 table

Variables:

YEARMONTH: double
EPISODE_ID: double
EVENT_ID: double
LOCATION: cell array of character vectors
LATITUDE: double
LONGITUDE: double
LAT2: double
LON2: double

Statistics for applicable variables:
```

	NumMissing	Min	Median	Max	Mean	St
YEARMONTH	0	202301	202306	202312	2.0231e+05	2
EPISODE_ID	0	175219	181491	187950	1.8156e+05	3.196
EVENT_ID	0	1062433	1107092	1154656	1.1092e+06	2.321
LOCATION	0					
LATITUDE	0	-14.3429	37.3509	66.6546	37.1230	5

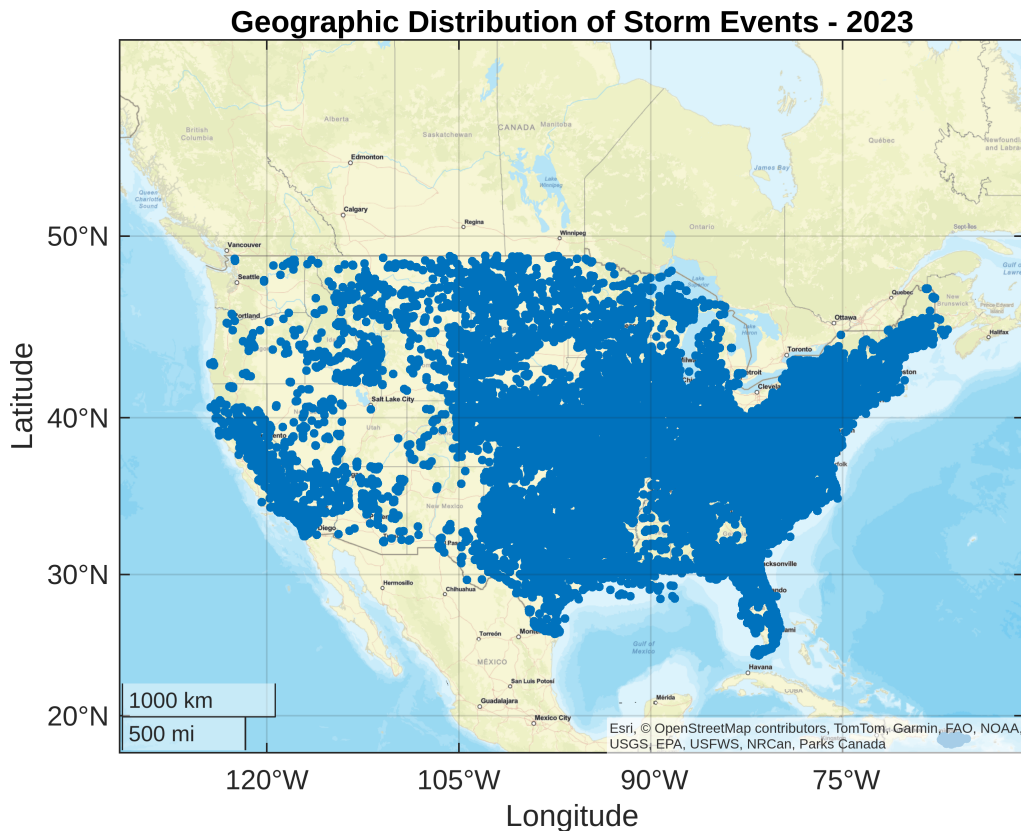
LONGITUDE	0	-170.8744	-88.7933	144.9095	-91.2688	15
LAT2	0	-1420574	3621000	6639276	3.1182e+06	1.359
LON2	0	-14454570	8514064	17052464	7.7004e+06	3.433

Basic Scatter Plot of Storm Locations

```
% Filter out rows with missing coordinates
validCoords = ~isnan(stormLocations.LATITUDE) &
~isnan(stormLocations.LONGITUDE);
stormCoords = stormLocations(validCoords, :);

% Create scatter plot of storm locations
figure
geoscatter(stormCoords.LATITUDE, stormCoords.LONGITUDE, 10, 'filled')
title('Geographic Distribution of Storm Events - 2023')
geobasemap streets

geolimits([17.4 58.8],[-131.5 -60.8])
```



Frequency Analysis Section

```
% Remove invalid coordinates
validLat = abs(paths.LATITUDE) <= 90 & abs(paths.LAT2) <= 90;
validLon = abs(paths.LONGITUDE) <= 180 & abs(paths.LON2) <= 180;
```

```
paths = paths(validLat & validLon, :);
```

```
% Remove missing or blank locations
```

```
stormCoords.LOCATION = string(stormCoords.LOCATION);  
validLocs = stormCoords(stormCoords.LOCATION ~= "", :);
```

```
% Count how often each location appears
```

```
[locCounts, locNames] = groupcounts(validLocs.LOCATION);
```

```
% Combine into a table
```

```
locationCounts = table(locNames, locCounts, 'VariableNames', {'Location',  
'EventCount'});
```

```
% Sort by count descending
```

```
locationCounts = sortrows(locationCounts, "EventCount", "descend");
```

```
% Show top 10 locations
```

```
topLocations = locationCounts(1:10, :)
```

```
topLocations = 10x2 table
```

	Location	EventCount
1	"SAN JUAN"	183
2	"PLAYALINDA BEACH"	75
3	"SPRINGVILLE"	63
4	"VIEUX CARRE"	58
5	"PONCE"	56
6	"MAYPORT NAS"	51
7	"NAGUABO"	49
8	"NORFOLK"	48
9	"FAIRVIEW"	46
10	"DUNLAP"	45

```
% Plot top 10
```

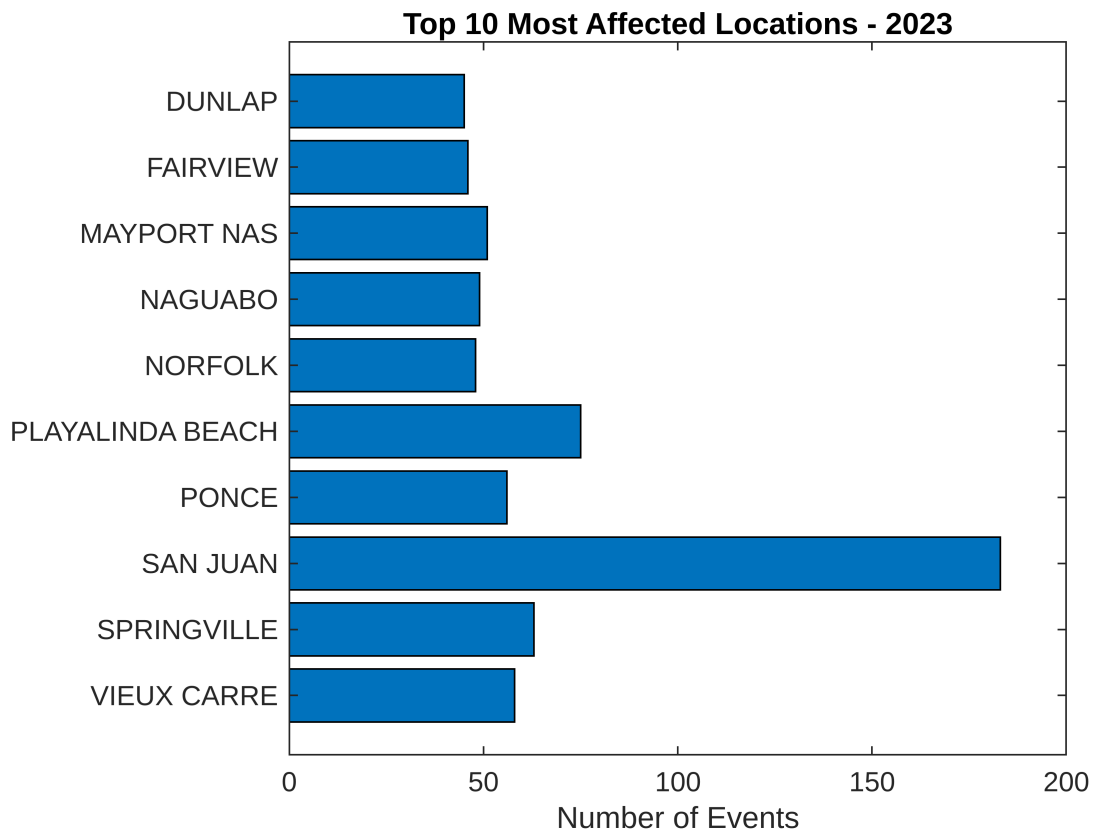
```
figure
```

```
barh(categorical(topLocations.Location), topLocations.EventCount)
```

```
xlabel('Number of Events')
```

```
title('Top 10 Most Affected Locations - 2023')
```

```
set(gca, 'YDir', 'reverse')
```



Summary

This analysis shows the geographic footprint of storm activity in the U.S. during 2023.